



<Code Club JUSL/>



DEPARTMENT OF INFORMATION TECHNOLOGY
JADAVPUR UNIVERSITY
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Google SWE Intern Internship Acceptance Guide

A complete guide for aspiring interns

Student's Experience & Suggestions

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Joyosmit Pal

Computer Science & Engineering

Name	Joyosmit Pal
Department	Computer Science & Engineering
Year of Grad.	2027

Experience

2 questions were asked in OA of moderate difficulty. 1 was based on graphs (something to do with components of graph, cant remember the exact details) and the other was based on just logic.

Interview round 1:

Q1) you are given a set of directed edges and vertices are from $[0, n-1]$. Also a set of locked vertices are given which cannot be entered or exited. Find the shortest path from a given source node to destination node. Follow question was-say you can now visit the locked nodes. Find the minimum number of locked nodes you have to visit in order reach destination from source. I solved them both using BFS and gave the time and space complexity. The interviewer was satisfied and asked if I had any questions for him. I asked about work-life balance at Google and what kinda projects do interns work with.

after about half an hour I got the mail for Round 2

Interview round 2:

You are given 2 sets $a=[x_1, x_2]$ and $b=[y_1, y_2]$ where these 2 represent the set of all real numbers r_1 and r_2 such that $x_1 \leq r_1 < x_2$ and $y_1 \leq r_2 < y_2$. Find the set difference (i.e., $a-b$) and their intersection. I coded the solution and explained time and space complexity. The interviewer was satisfied and asked if I had any questions for him. I asked about the career trajectory at Google and what will happen after the internship ends.

Prep. Tips

- 1) Know how to communicate your ideas to the interviewer
- 2) Take any hints you get from your interviewer and incorporate them into your solution.
- 3) Have some questions ready to ask them at the end of your interview

✓ **Selected**

2

Abhirup Pal

Computer Science & Engineering

Name	Abhirup Pal
Department	Computer Science & Engineering
Year of Grad.	2027

Experience

The drive begins with the OA round. It is a 1-hr round with 2 questions. Due to lack of time, I was able to only complete 1 problem completely, and for the other I had to come up with a solution that intentionally does not handle some cases but should finish most (if I tried to finish the second problem with a general solution, I would not have been able to submit it). Both the problems passed the visible test cases, but the second problem "should not" pass all the hidden ones. This round had questions from general graphs, DSU, DP, Segment trees and recursion. After a few days, the shortlist came. I was selected for the interviews.

I had my 2 interviews on 2 consecutive days. Both were mentioned to be eliminative.

In the first round, I joined and was asked to introduce myself. I did so. My interviewer asked about what project I was doing at that time, not necessarily the ones in my resume. I mentioned about the chess app I was building. This took about 5 minutes. Then, we went to the coding problem. The problem was a simple BFS problem with corrupted node handling. I did so. For the follow-up, I was asked to query about the distance in the graph and she added more conditions. I implemented it. Then she asked me, how I would find critical nodes which I should backup in case more nodes get corrupted. I explained her the process of Tarjan's algorithm. She looked satisfied. I asked if I had to code it, she said No. All this took 20 minutes. Since time was still left, she brought a 3rd question, this was also BFS, but I had to formulate the graph first with a condition (basically a BFS to make the graph) and then another BFS to solve. This was the 35th minute mark. After this, we were simply having a chat about life at Google. The interview was around 40-45 minutes.

A few hours after the interview, I received a mail from google saying that I progressed to round 2. And it was scheduled for the next day.

The next day, I joined the interview. There was no introduction. The interviewer asked my name and we straight away went for the coding question. The question was:

Given 3 sorted integer arrays a, b, c of length n each and an integer D, find the number of tuples (i, j, k) such that:

$|a[i] - b[j]| \leq D$
 $|b[j] - c[k]| \leq D$
 $|c[k] - a[i]| \leq D$

I mentioned that brute force of $O(n^3)$ of course exists. Since the array was monotonic, we could use binary search or two pointers. If we fix the array with the smaller number (say a) and create windows in b and c for the permissible indices, we can simply multiply number of indices in those windows; then do the same keeping b as the smaller, followed by c. (Noticing the pointer to the middle value of the triplet is irrelevant, just the difference between largest and smallest needs to be $\leq D$) There was some overcounting in the smaller side for handling the fixing of smaller array, which I noticed while coding itself and fixed. The final solution was $O(n)$ time and $O(1)$ space. It had been 35 minutes of the interview.

The interviewer asked me to convert the fixing of the 3 cases (a, b or c first) into a single function. I did so. I used an increment trick to handle the asymmetries of the 3 fixed to take into account overcount of cases where $a[i] = b[j]$ or $b[j] = c[k]$ or $c[k] = a[i]$ in fixing the lower bound. The interviewer asked me to explain why the two-pointers/sliding-window approach gave $O(n)$ time. I explained the concept of amortization. The interviewer seemed satisfied. There was hardly any time left. We chatted for a couple of minutes, and the 45 minutes was over.

The next day, in the afternoon, I received a message saying I was shortlisted.

Prep. Tips

Start with DSA early. This is more-or-less the only thing questioned in the interviews. Cover all the topics and implement them. The questions are never "implement quicksort" but something that requires you to first come up with the algorithm.

Also, the interviews are taken in a text editor made by google with no autocomplete, not even bracket completion. So, I would recommend, before the interview practice coding in notepad or google docs.

Addit. Notes

Do not panic. The interviewers are very friendly. You can ask them about anything you want. If you are unfamiliar with a topic tell them. Often times, they might even give hints.

✓ **Selected**

3 Sayan Dutta Information Technology	
Name	Sayan Dutta
Department	Information Technology
Year of Grad.	2027

Experience

In the OA there were 2 DSA questions and we had 60 minutes to solve it. The first question was based on graph and can be solved using connected components only. the second question was given a integer n, find number of palindromes $\leq n$ such that ith digit occurs i number of times (like if 2 occurs in the palindrome it should occur exactly 2 times). It could be solved by generating all possibilities using bfs but during the OA i was not able to fully optimize the solution. The question was for n upto $1e17$ and my solution was correct for n upto $1e13$ only

The interview took place on Google Meet, and we used Interview Docs (similar to Google Docs but had syntax highlighting for code). The interviewer began by introducing herself and then asked me to introduce myself. Then she verbally explained the question. The problem was "given a undirected unweighted graph and a list of nodes which contains the nodes with a data center, my task was to find a node in the graph in which if we make a new data center then it will be furthest from all the other data centers" - initially the question felt a bit ambiguous so I asked the interviewer for clarification and came to a conclusion that the question was simply find a node such that the minimum distance of it from any of the data centers is maximum. Then I explained my brute force and its time complexity and she asked me to optimize it, the optimal solution was using multisource bfs from all the initially given data centers to find our answer. I told her the optimal solution and she asked me to dry run it and finally to code it. Then she followed it up with the question "what if the graph was weighted instead", the solution to this was to use multisource dijkstra instead of bfs, I told her that and finally coded up this solution as well, the interview ended in about 30 minutes and about 15 minutes later i got the email for my second round

The second round was easier than the first round, the interviewer told me to introduce myself and then she pasted a question (she was trying to observe what is my approach when i see a question, so she asked me to go through the question myself and if i have a doubt then clarify it) the first question was given a necklace (linear) of $2*r$ rubies and $2*d$ diamonds find 2 or less partitions such that we can distribute the parts of the necklace between two thieves and both of should get d diamonds and r rubies. it could be solved using a sliding window of size $r+d$. The next question was given an array which contains start_time of some processes and a time t which each process takes to complete, what is the

minimum time to complete all the processes if our computer can handle c processes simultaneously.it could be solved using queues. I told the optimal solutions ,dry run them on given test cases, wrote the code and mentioned time and space complexity for both .Even though I got the space complexity wrong on the second question, the interviewer was helpful and she guided me through the mistake. Overall the round was of 45 minutes.

Prep. Tips

Graphs and Trees are most important , you need to have a strong command over bfs, dfs and their usage in different cases. you need to be comfortable dry running code (not only with pen-paper ,but you have to explain someone verbally how the code is working for a certain tc) and you have to speak whatever you are thinking ,then only interviewers will be able to help

✓ **Selected**

4 Somnath Chattaraj Information Technology	
Name	Somnath Chattaraj
Department	Information Technology
Year of Grad.	2027

Experience

In the online assessment, there were two questions: one on dynamic programming (DP) on string and another that was more challenging, requiring knowledge of segment trees. I successfully solved the DP question. For the second one, I implemented a brute force solution that passed the given test cases.

In the first interview, I was tasked with a room allocation problem. The goal was to allocate rooms such that most interns got a room, considering that some interns wanted housemates and others didn't. The input was provided as a vector of structs, and while I was able to come up with a solution, there were a number of edge cases that added complexity. Although my code missed a few edge cases, the interviewer was understanding and gave positive feedback after I performed a dry run on one of the edge cases.

The second interview involved an array problem where the objective was to find the maximum subarray sum. The catch was that if $arr[i] == arr[j]$ for $i \leq j$, I had to compute the sum of the subarray between indices i and j . Initially, I proposed a brute force solution involving multiple loops. However, when the interviewer asked me to optimize it, I quickly modified the Kadane's algorithm with a hash map to come up with an efficient solution. The interviewer was pleased with the approach and didn't ask for a dry run. The interview wrapped up in 20 minutes.

Prep. Tips

Striver A-Z and SDE sheet should be on tips.

✓ **Selected**

Name	Srijani Roy
Department	Information Technology
Year of Grad.	2027

Experience

I secured this opportunity through an **off-campus** process, without any referral involved.

After my resume was shortlisted, I was contacted by a recruiter. This began with an initial phone screening, where the recruiter asked me to give an introduction. The conversation then moved on to a few background questions to assess my preparation and interests. These included how many DSA questions I had solved, which coding platforms I used, and what my CGPA was. The recruiter also asked about my primary programming language, as well as the tech stack I had used in my projects. I was asked to give a brief overview of one of my projects, including the technologies involved and the overall goal.

The call concluded with the recruiter asking for my availability so that she could schedule the first round of technical interviews.

Both interview rounds were scheduled on short notice, and I had to manage them alongside my ongoing semester exams. In fact, one of the rounds took place between two of my exams, which added to the pressure. Moreover, I'm sharing this experience after quite some time has passed, so I don't remember the exact questions word-for-word.

Each interview round focused entirely on Data Structures and Algorithms. Once the recruiter confirmed the date and time, I received a Google Meet link along with a shared interview document. The document is a plain, blank file with no pre-written code. During the interview, the interviewer pastes the question in front of you, and you're expected to write the code from scratch.

Technical Round 1

The first technical round lasted for about 45 minutes. It started with a brief self-introduction, which I kept short in order to make the most of the available time. We immediately proceeded to the first question, which was based on trees. I initially discussed the brute-force approach with the interviewer, followed by the code implementation. After that, I provided an optimized version and explained that as well.

The second question was based on graphs, and followed a similar pattern of discussion, coding, and optimization. For both questions, I was also asked to calculate the time and space complexity of my solutions. I also had to dry run the questions to demonstrate their correctness. At the end, the interviewer gave me the opportunity to ask any questions I had. Since I was quick with coding, the round was completed efficiently within the allotted time.

The next day, the recruiter informed me that I had cleared Round 1 and was selected for Technical Round 2.

Technical Round 2

The second technical round was more intense and lasted for around 55 minutes, slightly longer than the usual 45-minute format. The interviewer asked me three questions, covering the topics of recursion, backtracking, DP and strings.

For each question, I discussed both the brute-force and optimized solutions, implemented the code and clearly explained my thought process. I was also asked to compute the time and space complexity for all the solutions. Since I had to work through three questions along with both brute-force and optimized approaches, the interview extended beyond the usual time.

Final Result

After waiting for over 1.5 months, I was informed that I had been selected.

Prep. Tips

Have some strong projects on your resume, along with your coding and educational achievements.

Practice dry running your solutions to ensure they work correctly.

Learn how to calculate the time and space complexity of your solutions.

During the interview, actively interact with your interviewer and try to clearly explain your code and thought process.

Additional Notes

For each coding question, try to ask clarifying questions to ensure you fully understand the problem.

Try to write comments in your code.

Use meaningful variable and function names to make your code clean and easy to understand.

✓ **Selected**
